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Text measurement analysis of green and low-carbon industrial policies through machine learning: insights from the Chengdu–Chongqing economic circle

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Text measurement analysis of green and low-carbon industrial policies through machine learning: insights from the Chengdu-Chongqing economic circle

Abstract: Green and low-carbon industrial policies serve as a key guideline for China's pursuit of its dual-carbon goals. Green and low-carbon industrial policies therefore play a crucial role in advancing green technology and fostering the sustainable transformation of traditional industries. This study examines the Chengdu-Chongqing economic circle as a case study, employing a machine learning model and the policy modeling consistency index to assess the intrinsic value, evolutionary characteristics and policy effectiveness of green and low-carbon industrial policies within the region. Findings reveal the following: (1) Three categories of policy objectives and eight types of policy instruments were identified; (2) Although the number of policy objectives is similar, both Chengdu and Chongqing primarily rely on demand-side policy instruments; (3) Over the three developmental stages of green and low-carbon industrial policy, policy priorities have slowly shifted from theoretical guidance to implementation control; (4) Although the policy effectiveness in Chengdu is slightly lower than in Chongqing, it exhibits greater stability. Enhancing policy level could substantially improve effectiveness; (5) Notably, whilst demand-side instruments are the most frequently employed in the Chengdu-Chongqing economic circle, our analysis of three policy centers indicates a reliance on both supply-side and environmental instruments, a pattern that may be influenced by China's unique policy landscape. This study not only provides policy recommendations for policymakers but also expands the field of industrial policy evaluation in the carbon-neutral era.

Keywords : Policy evaluation; Machine learning; PMC-Index model; Green and low-carbon industrial policy; Chengdu-Chongqing economic circle

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1. Introduction

Green and low-carbon industrial policies (GLIPs) represent a distinct subset within industrial and environmental policy frameworks, designed to facilitate the transition to sustainable energy systems and support long-term carbon neutrality (Liu & Peng, 2022). As an emerging policy category, GLIPs are defined by their emphasis on accelerating the adoption of clean energy and promoting the green transformation of industrial structures (Li et al., 2025). In recent years, these policies have attracted growing attention as nations seek to reconcile economic development with environmental sustainability.

The large-scale exploitation of fossil fuels has led to severe environmental issues, including increased carbon dioxide emissions, resource depletion and global warming (Waheed et al., 2019). According to the International Energy Agency (IEA), global energy-related CO_2 emissions increased by 1.1% in 2023, rising by 410 million tons to a record high of 37.4 billion tons (IEA, 2023). Addressing global warming requires a substantial reduction in fossil fuel consumption and a greater share of renewable energy in the global energy mix (Cho et al., 2023; Fawzy et al., 2020; Su et al., 2025). The transition to clean energy necessitates industrial transformation (Guchhait et al., 2024; Sarkar, Ganguly, et al., 2024; Su et al., 2024), prompting both developed economies and China to accelerate clean energy deployment (IEA, 2024). Several countries and regions have introduced policies to encourage the adoption of clean energy technologies, including the EU's Fit for 55 (2021), Japan's Green Growth Strategy for Achieving Carbon Neutrality by 2050 (2021) and China's Opinion on Accelerating the Comprehensive Green Transformation of Economic and Social Development (2024). A systematic study of GLIPs is essential for optimising industrial resource allocation and ensuring the sustainable development of global energy systems (Li et al., 2025; Plötz et al., 2023).

As a critical component of public policy management and optimisation, policy evaluation involves analysing and assessing both the historical causes and future impacts of policies on society using various theoretical frameworks as well as quantitative and qualitative models (Estrada, 2011). Policy review not only determines a policy's intrinsic value but also assesses the effectiveness of its formulation and implementation, guiding decisions on whether to continue, improve or terminate the policy (Hildén, 2011; Huitema et al., 2011; Sanderson, 2002). Since China introduced its dual-carbon goals, an urgent need has been raised for a comprehensive evaluation of the value and effectiveness of newly implemented GLIPs to assess their role in

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supporting industrial development (X. Zhang et al., 2024). However, the inherent complexity of policies poses a remarkable challenge to policy evaluation (Bandelow et al., 2022). Although numerous studies have examined industrial sustainability policies through qualitative analysis (P. Bianchi et al., 2021; Ferrannini et al., 2021), researcher subjectivity can lead to inconsistencies in findings. Additionally, qualitative methods are often inadequate for analysing large-scale policy texts (Patrizio Bianchi et al., 2021). To address these challenges, this study proposes a policy evaluation framework based on machine learning and the policy modeling consistency-index (PMC-Index) model using GLIPs as a case study. This research is a case study of the Chengdu-Chongqing economic circle (CCEC), one of China's major urban agglomerations, with an aim to address the following three research questions (RQs):

RQ1: What are the objectives of GLIPs, and which instruments does it employ to achieve them?

RQ2: How have the objectives and instruments of GLIPs evolved throughout the development of the industry?

RQ3: How can the effectiveness of GLIPs be assessed, and what are its key advantages and disadvantages?

To address the RQs, we conducted an in-depth investigation of GLIPs. A literature review was first undertaken to identify existing research gaps and inform the development of potential research directions. To answer RQ1 and RQ2, this study employed the K-means clustering algorithm and the latent dirichlet allocation (LDA) clustering model to analyse the dynamic changes in policy instruments and objectives. Using the PMC-Index, we quantified the policy effectiveness of GLIPs and highlighted its advantages and disadvantages, thus answering RQ3. The comprehensive analysis suggests that optimizing GLIPs for the CCEC area involves balancing environmental and supply-side policy instruments, restructuring industrial policy, and enhancing the effectiveness of policy instruments. This paper underscores the innovation and contribution of our study in the following three areas: (1) We propose a machine learning-based framework for identifying policy objectives and instruments, overcoming the limitations of traditional policy evaluation methods in large-scale text processing; (2) We visualise the advantages and disadvantages of policies through comprehensive multidimensional indicators, thus resolving the 'black box' issue in policy effectiveness assessment; (3) Our research fills the gap in studies on the application of policy instruments at the city level (Yang et al., 2021). The findings not only provide a novel

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methodological approach for optimising GLIPs in China's CCEC but also offer insights for promoting more proactive and effective sustainable development policies in regional and urban clusters.

The remaining sections are arranged as below. Section 2 details the literature review. Section 3 presents the methodology. Section 4 exhibits the results and analysis. Section 5 draws the discussions. Section 6 summarises conclusions.

2. Literature review

2.1 Policy evaluation

Policy evaluation is a critical component of policy implementation (Edler et al., 2008). Through systematic and scientific evaluations, governments and relevant institutions can mitigate decision-making biases stemming from empiricism whilst ensuring the long-term sustainability of policies (Mavrot et al., 2025). Furthermore, transparent policy evaluations enhance the credibility of governance (Kotzampasakis & Woerdman, 2024). Consequently, a comprehensive and objective assessment of policy texts not only advances empirical research in this field but also plays a vital role in elucidating the decision-making process (Mundaca et al., 2022). As public policy research progresses, the study of policy semantics and the evaluation of policy effectiveness have emerged as the predominant paradigms for policy assessment (Goyal et al., 2022). However, existing studies often remain fragmented. This suggests that future research should integrate these dimensions to achieve a more holistic understanding of policy performance.

2.2 Policy semantic research

Policy semantic research is a crucial method for understanding the core focus of policies (Zacharewicz et al., 2024). Existing studies, grounded in interpretivism, constructivism and positivism, can be broadly categorised into three approaches. Early research predominantly focused on interpretivism and constructivism. Interpretivism emphasises the subjective interpretation of policy implementation. For example, Zulu et al. (2013) assessed the policy development process of community health assistants in Zambia using interview analysis (Zulu et al., 2013). By contrast, constructivism examines how policies shape social realities, with critical discourse analysis (CDA) serving as a core approach. Liasidou (2008) applied CDA to evaluate Cyprus's inclusive education policy (Liasidou, 2008), whilst Coffey and Marston (2013) explored how neoliberal and ecological modern discourses combined to influence environmental policy in

Victoria, Australia (Coffey & Marston, 2013). Although these studies shed light on policy power structures and ideological biases, they may be subject to researcher interpretation, potentially distorting findings. To address this concern, positivist approaches centered around content analysis, began to emerge. Content analysis, utilising statistical methods, offers a more objective and systematic way to analyse policy texts. For instance, Takahashi et al. (2007) conducted a comparative analysis of high-frequency and subject-specific words in policy texts (Takahashi et al., 2007), whilst Liu et al. (2024) employed social network analysis to examine word co-occurrence in policy documents (Liu et al., 2024). These methods persistently face challenges in handling large-scale policy texts due to slow processing speeds. Nevertheless, a critical gap persists: interpretive, constructivist, and positivist methods are often treated as mutually exclusive, while their complementarity remains underexplored. Recently, the use of text mining, such as natural language processing (NLP) technologies, has emerged as a cutting-edge approach for policy text analysis (Ahn et al., 2021; Davidson et al., 2025; D. D. Zhu et al., 2023). A synthesis of qualitative insights with NLP-based large-scale analyses could reconcile depth and breadth, offering more robust semantic evaluations of policy texts (Kim et al., 2019).

2.3 Policy effectiveness measurement

Policy effectiveness measurement is another core issue in policy evaluation research, with its aim to quantify the actual impact of policy implementation through scientific methods to ensure the effective realisation of policy objectives (Estrada, 2011). Existing studies divide policy effectiveness measurement into two categories. The first is testing the significance of a certain indicator to determine the actual effect of policy implementation. This type of research is represented by econometric models, including the differential-in-differences model (Cellini & Cuccia, 2024) and the synthetic control method (de Boer et al., 2022). The second category is judging the policy terms themselves. A study combined text subject modeling with Delphi method to evaluate circular economy policy (Mahanty et al., 2021), whilst another research conducted a case study of Slovenian food and nutrition policies through semi-structured interviews (Kugelberg et al., 2012). The policy modeling consistency index proposed by Estrada in 2011 puts forward a new research direction for quantifying the effectiveness of policy texts (Estrada, 2011). PMC-Index model allowing researchers to bridge the gap between text-based evaluations and outcome-based measurements. It integrates the qualitative and quantitative approaches of traditional policy evaluation, leveraging the strengths of both while mitigating their respective limitations (Table 1).

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Table 1. Comparison of traditional qualitative, quantitative policy evaluation methods and PMC-Index model

Method Type/Model	Strengths	Limitations	References
Qualitative policy evaluation methods	Provides in-depth understanding of policy context and implementation; captures complex mechanisms	Difficult to quantify results; subjective; prone to researcher bias	(Liasidou, 2008; Fisher & Hamer, 2020)
Quantitative policy evaluation methods	Produces objective, reproducible measures; suitable for statistical analysis and modeling	May overlook policy context and implementation details; struggles to capture non-quantifiable factors	(Cellini & Cuccia, 2024; Oosthuizen & Inglesi-Lotz, 2022)
PMC-Index model	Combines the strengths of qualitative and quantitative methods; quantifies policy text features and links them to outcomes; supports cross-policy comparison	Requires high-quality textual data and clear coding rules	(Estrada, 2011; Song et al., 2024)

2.4 GLIP research

Traditional industrial policies often rely on mandatory measures to promote sustainable industry development, but this approach can result in a lack of market-driven momentum (Abbott & Fitjar, 2025). Although environmental protection policies, such as those informed by Porter's hypothesis, offer some planning guidelines, numerous studies have demonstrated that the tension between economic growth and environmental degradation cannot be alleviated by a single environmental governance policy (Wang et al., 2019). These challenges have prompted the government to reconsider and reassess its policy approach. Qin et al. argues that policies should be formulated to align with the developmental context of a specific country or region, aiming to balance industrial growth with environmental protection (Qin et al., 2021). These studies primarily focus on macro-level scales, such as national and provincial scales

The term GLIPs is proposed against the context of China's distinctive environmental protection development strategy (Li et al., 2023). Operationally, GLIPs typically encompass measures such as industrial carbon emission standards, fiscal incentives for green manufacturing, investments in clean energy infrastructure, and support for low-carbon technological innovation (Han, 2016; Li et al., 2025; Xie et al., 2018). Incorporating international concepts such as sustainable development and green technology, GLIPs have broad applicability in other countries and regions (Mundaca & Markandya, 2016). However, unlike traditional environmental protection

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policies, GLIPs encompass extensive areas, which increases the complexity of both their formulation processes and textual content. This complexity not only lies in the wide policy scope, but also in the challenges of inter-departmental coordination, regional adaptability, and the ambiguity of policy language, making the evaluation of GLIPs inherently more difficult than traditional industrial or environmental policies.

2.5 Literature gaps

This study aims to address the following three current literature gaps:

Firstly, the rapid growth of policy data necessitates adaptable and forward-thinking text data processing methods. Traditional policy evaluation relies heavily on subjective inferences, leading to uncertain or heterogeneous results (Song et al., 2024). Few studies have explored the application of text mining models, such as NLP, in the semantic analysis of policy texts. Furthermore, few studies have integrated NLP models to develop a comprehensive framework for policy text analysis. By contrast, most research continues to rely on case studies or basic statistical models to assess policy semantics.

Secondly, despite the complex assumptions and tests in econometric models, researchers often struggle to control for and evaluate the impact of external factors on policy effectiveness. These models also fail to address the 'black box' problem of policy effects. A more direct approach, which evaluates policies themselves, allows for a clearer assessment of their value (Christie & Lemire, 2019). Additionally, existing studies often lack a structured policy effectiveness evaluation database for assessing the impact of policies in specific dimensions.

Thirdly, Chinese GLIPs have developed within a unique political, economic, cultural and social context (T. Yang et al., 2023). Unlike other environmental or industrial development policies, fully quantifying and evaluating GLIPs entail complex and challenging processes. Consequently, explaining the intrinsic characteristics and mechanisms of GLIPs remains difficult. The traditional environmental policy assessment system may be inadequate for evaluating GLIPs. Additionally, few studies focus on urban areas or propose targeted policy formulation strategies. For instance, Yang et al. (2021) argue that, unlike national policies, local policy development can be tailored to support the growth of the local new energy vehicle industry, enhancing its relevance and effectiveness (Yang et al., 2021). Similarly, Boussalis et al. (2018) suggest that, compared to federal governments, local governments (cities) can implement more impactful measures to address climate change (Boussalis et al., 2018). Assessing environmental policies at the urban scale,

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therefore, facilitates the formulation of measures that are better aligned with local needs. Consequently, there is a pressing need to establish a comprehensive policy evaluation system for GLIPs issued by the CCEC and to examine its applicability across various regions and contexts.

To address these gaps, this study employs an integrated machine learning approach that combines the K-means clustering model, LDA clustering model and PMC-Index model. This approach enables quantitative and visual analysis of policy texts, providing a logical explanation of the evolution and implementation effects of GLIP. By overcoming the high subjectivity and extensive labor involved in traditional policy text research, the proposed framework offers accurate and scientific guidance for industry development, facilitating the formulation of effective policies.

3. Methodology

3.1 Research framework

This study develops a policy text mining framework incorporating the K-means clustering model, LDA clustering model and PMC-Index model to identify policy content and assess policy effectiveness. The empirical analysis focuses on CCEC, and the research steps are as follows (Fig. 1): Firstly, we deconstruct the connotation of GLIPs and select the policy instrument types proposed by Roy Rothwell and Walter Zegveld. Based on this process, we encode the policy text to classify each GLIP under its corresponding policy instrument type. This systematic coding ensures that the policy text aligns with the requirements for constructing the analysis framework. Secondly, we integrate the LDA and K-means clustering models to create an X-Y two-dimensional analysis framework for identifying policy text content. The K-means model identifies policy schemes under policy instruments (X dimension), whilst the LDA model identifies policy objectives (Y dimension). Thirdly, we select key policy events to divide GLIPs into three stages. Using the framework from the second step, we analyse the characteristics of policies in each stage. Finally, we apply the PMC-Index model to assess selected GLIPs. After calculating the PMC score and generating the PMC-Surface map, we visualise the strengths and weaknesses of the policy text. The study concludes with a discussion of the data analysis results and corresponding policy recommendations.

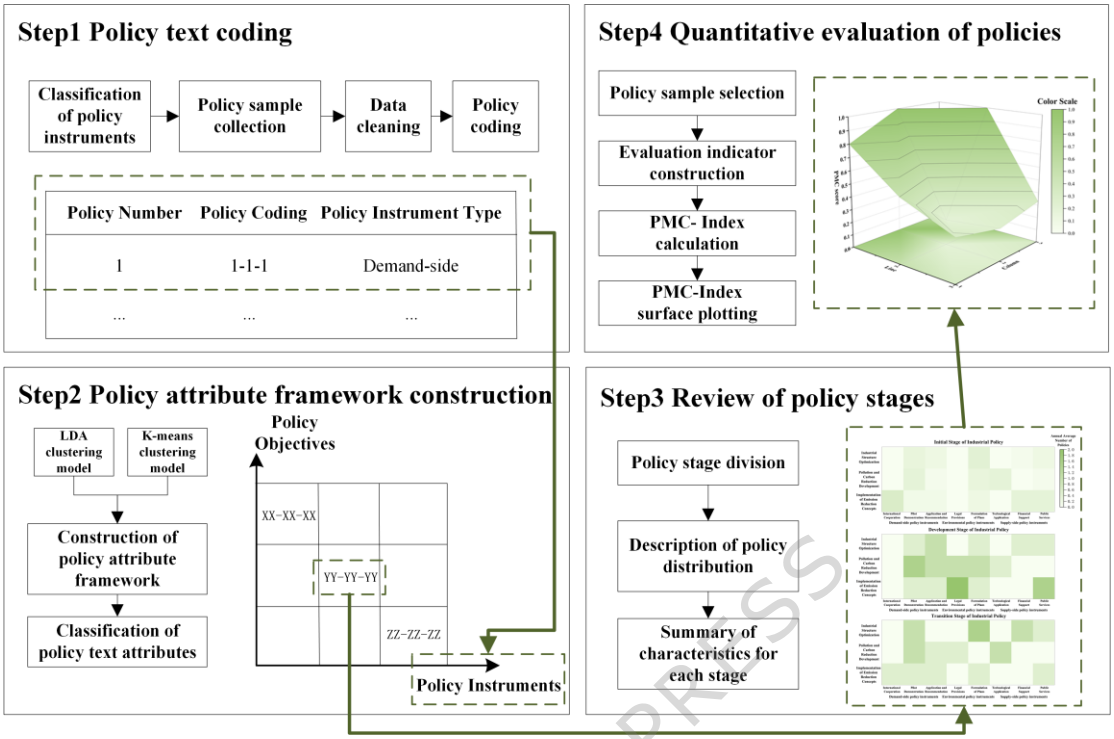


Fig. 1 Technical route of research design

3.2 Study area and data collection

3.2.1 Study area

To optimise the energy structure and promote sustainable development, the Chinese government has introduced the concept of green and low-carbon industries (GLI) as a key national strategy (T. R. Yang et al., 2023). The CCEC is one of the largest metropolitan regions in western China (Fig. 2), contributing 6.5% of the national GDP and 30.4% of the western region's GDP (2023). CCEC has a vast industrial base, with the output value of its electronics, automotive, equipment manufacturing and consumer goods industries exceeding one trillion yuan. Additionally, CCEC serves as a key pilot region for China's GLIPs (Gou & Liu, 2022; Li et al., 2025). In February 2023, the Chinese government released the Planning Outline for the Construction of the CCEC, underscoring the critical role of GLI in shaping industrial development. Similar to other industrialised regions worldwide—such as the Rust Belt in the United States (Atkins, 2024), the Ruhrgebiet in Germany (Henninger, 2008) and the Delhi-Mumbai Industrial Corridor in India (Garg et al., 2002)—CCEC faces challenges related to rising greenhouse gas emissions. Evaluating the effectiveness of CCEC's policy implementation not only provides insights into optimising industrial energy consumption and resource allocation in China but also offers guidance for

sustainable industrial development in other city clusters globally (Bhattacharya et al., 2024). Furthermore, assessing the policy impact of CCEC aligns with the United Nations' sustainable development goals, particularly Goal 7 (affordable and clean energy), Goal 9 (industry, innovation and infrastructure), Goal 11 (sustainable cities and communities) and Goal 13 (climate action) (UN, 2015). The findings of this study contribute valuable data for regional sustainability assessments. Therefore, this study evaluates the implementation effectiveness of GLIPs in CCEC.

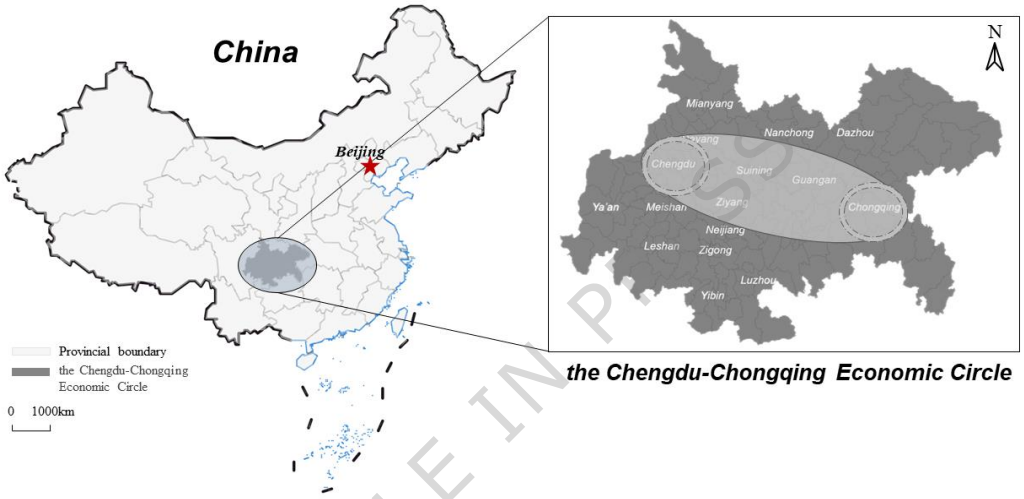


Fig. 2 Geographical location of CCEC

3.2.2 Data collection and cleaning

To ensure the representativeness and reliability of the selected policies, this study first collected policy text data from official Chinese government websites and authoritative policy databases. The following is an explanation of the source, scale, time range, and preprocessing of policy texts. Data were gathered from the Peking University PKULAW Database and government websites of the Sichuan and Chongqing provinces. Policies related to GLIPs were identified using key search terms such as 'green industry', 'low carbon industry' and 'green and low-carbon industry'. A total of 186 policies from Chengdu (60.78%) and 120 policies from Chongqing (39.22%) were initially collected (2007-2023). The collected policy texts were then manually cleaned. After removing duplicates and irrelevant content, 54 policies from Chengdu (52.94%) and 48 policies from Chongqing (47.06%) were selected, totaling 188,934 characters. Table S1 and Table S2 in the supplementary materials provide an overview of the policy text data.

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3.3 Research design

3.3.1 Type of GLIP instruments

Defining policy instruments is a fundamental prerequisite for policy semantic analysis (Burke & Stephens, 2017). Selecting appropriate policy instruments enables researchers to assess how policies address public issues and evaluate their role within the broader policy system (Mundaca & Markandya, 2016). Consequently, the selection of policy instruments is a critical topic in public policy research.

Advancements in public policy research have led to the development of various classifications of policy instruments. E.S. Kirschen identified 64 general policy instruments but did not systematically organise them or theorise their origins and effects. Howlett and Ramesh categorised government instruments into voluntary (non-mandatory), mandatory and mixed instruments based on the degree of government intervention (Howlett & Ramesh, 2003). Linder and Peters proposed a broader classification, including command clauses, financial assistance, regulation, taxation, exhortation, authority and contracts (Linder & Peters, 1989). Whereas these classification frameworks provide valuable insights, they are not specifically designed to guide industrial development and thus lack comprehensive applicability in this context.

GLIPs contain a regulatory framework centered on sustainable development, integrating both industrial growth principles and environmental sustainability. It aligns with the policy instrument classification proposed by Roy Rothwell and Walter Zegveld. Notably, their framework has been widely applied in policy analyses related to energy reduction and sustainable industrial development, particularly in China (Dong & Liu, 2020; Xu et al., 2024; Zhang et al., 2020). This scenario demonstrates the applicability of their classification system to GLIP analysis.

Accordingly, we divide policy instruments into supply-side policy instruments, environmental policy instruments and demand-side policy instruments, with their primary connotation and definition shown in Table 2.

Table 2 Meaning of Roy Rothwell and Walter Zegveld's policy instruments

Policy instrument dimension	Focus on key points	Policy instrument definition
Supply-side policy instruments	Technological progress	The government provides the elements the market players need to promote scientific and technological research and development.
Environmental policy	Environmental	The government regulates the business behaviour of market

instruments	impact	entities and creates an environment for innovation.
Demand-side policy instruments	Market role	The government buys innovative products from market players to boost market demand.

Supply-side policy instruments promote the development of GLI, which primarily means that the government supports industrial research and development from the supply side and supplies elements to promote industrial development. The effect of this policy is relatively infinite, but it will require some time to take effect because industrial research and development typically requires a long time. Environmental policy instruments support or influence the development of GLI. The government can implement such policy instruments by implementing ecological protection policies and formulating industrial development plans. Environmental policy instruments are similar to Supply-side policy instruments, which also need some time to achieve results, but their effects are relatively durable. Drivers of development have been the GLI Demand-side policy instruments, which focus on driving technological innovation and new product development by actively exploring and stabilising the latest technology application market to promote industrial development. Compared with the first two policy instruments, the effect of demand-side policy instruments can appear faster. Generally, it can only play a one-time stimulus effect, and the effect may not be as lasting as the first two policy instruments (Fig. 3).

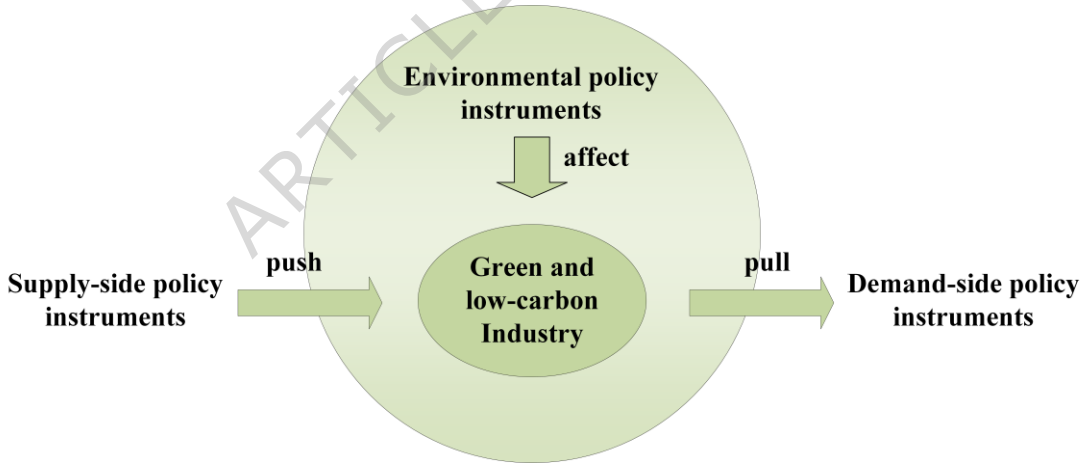


Fig. 3 Role of policy instruments in GLI

After identifying the three categories of policy instruments, the policies can be coded to determine their respective instrument types.

3.3.2 Coding of policies

Policy text coding is a text research paradigm based on content analysis for policy structure processing and quantitative analysis, which typically includes determining the analysis dimension,

defining the analysis unit, reliability and validity testing, frequency statistics and dimension analysis, to name a few. The non-quantitative information of policy texts leads to the lack of operational policy instrument classification standards, which renders the research and judgment of policy evolution and development trends less convincing. It also results in a lack of foresight and guidance for policy practice. Therefore, before policy text coding, establishing a suitable coding system and framework is first necessary (Wang & Xing, 2022). Based on the types of policy instruments, we encode the text in order of policy number, chapter number and article number (Table S3). Table 3 summarises the types and sources of policy instruments for GLIPs in the CCEC.

Table 3 Number of policies under each type of policy instrument

Types of policy instruments	Policy quantity	Policy source
Supply-side policy instruments	15	Chengdu
	16	Chongqing
Environmental policy instruments	16	Chengdu
	12	Chongqing
Demand-side policy instruments	23	Chengdu
	20	Chongqing

To ensure the accuracy of quantitative analysis after the policy text is encoded, the paper conducts a reliability test on the encoded text. This paper calculates coding consistency based on Holsti's Coefficient Reliability in content analysis. When the coding consistency coefficient of different coders is more significant than 0.8, it indicates that the coding result of the policy document is good and can pass the reliability test. Equations 1 and 2 present the calculation of coding consistency by Halsty's formula. Among them, K represents the coding consistency coefficient, n represents the number of coders, M represents the number of identical clauses encoded by coders and the number of codes of n coders is represented by N_1 to N_n. R represents coding reliability. A larger R means that the higher the coders have in terms of the degree of consistency with the results obtained by policy coding, the higher the credibility.

$$K = \frac{nM}{N_1 + \cdots + N_n}, \quad (1)$$

$$R = \frac{nK}{1 + \cdots (n-1)K}. \quad (2)$$

After coding the system, we proceed with the structured analysis of the policy text. Following standardisation, text mining techniques are applied to interpret the policy's underlying meaning.

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3.3.3 Policy text mining model

This paper employs a text mining model to construct an X-Y two-dimensional analytical framework, enabling an examination of governmental methods and targeted issues. The encoded policy text is then categorised within this framework. By integrating text mining techniques, we enhance traditional qualitative policy analysis, improving both efficiency and accuracy in processing policy text data.

For the X-dimension, we utilise the K-means clustering model to classify sub-dimensions of policy instruments, thereby constructing the policy instrument dimension. Originally introduced by Steinhaus in 1955, K-means was initially applied to numerical data clustering (Jain, 2010; Kanungo et al., 2002). Owing to its high computational efficiency and strong performance in high-dimensional data clustering, it has since been widely adopted across various fields.

Compared to alternative methods, K-means offers a more stable and scalable approach for dynamic policy text analysis. It employs hard clustering, assigning each document to a single, clearly defined category—an approach well suited for the explicit classification of policy instruments. For example, although DBSCAN is effective in detecting clusters of arbitrary shapes, its performance declines when applied to high-dimensional and sparse datasets such as textual data, and it is notably sensitive to parameter selection. Consequently, K-means has been extensively applied in policy text clustering (Abad, 2014; Allen et al., 2017; Seo et al., 2020; Song et al., 2022; D. D. Zhu et al., 2023). Ibrahim et al. (2018) demonstrated that K-means outperforms DBSCAN in text clustering, exhibiting higher term precision, term recall and topic recall across different clusters. Despite the varying performance of clustering algorithms such as B-ngram and document pivot clustering, K-means is computationally more efficient (Ibrahim et al., 2018).

For a given dataset $X = \{x_1, \dots, x_i, \dots, x_n\}$ consisting of K d-dimensional data points ($X \in R^d$), the k-means clustering model partitions the data into K clusters, denoted as $C = \{c_k, i = 1,2, \dots, K\}$. Each cluster c_k is associated with a centroid μ_k. Euclidean distance is used as the similarity measure, and the objective function minimises the within-cluster sum of squared distances between each data point and its corresponding cluster centroid μ_k, as expressed in Equation 3.

$$J(c_k) = \sum_{x_i \in C_k} \|x_i - \mu_k\|^2. \quad (3)$$

The clustering objective is to minimise the total sum of squared distances, expressed as $J(C) = \sum_{k=1}^{K} J(c_k)$ (Equation 4).

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$$J(C) = \sum_{k=1}^{K} J(c_k) = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 = \sum_{k=1}^{K} \sum_{i=1}^{n} d_{ki} \|x_i - \mu_k\|^2, \quad (4)$$

where $d_{ik} = \begin{cases} 1, x_i \in c_i \\ 0, x_i \notin c_i \end{cases}$. Clearly, based on the least squares method and the principle of lagrange multipliers, the cluster center μ_k should be the average of the data points in class c_k. The clustering process converges when the mean value remains unchanged after an iteration.

For the Y-dimension, this study employs the LDA clustering model to extract policy topics and construct a policy objective chain. LDA refers to the latent dirichlet distribution, a text data analysis model based on the PLSA model often used in text topic analysis. In the LDA model, text is seen as a collection of topics, each of which is a collection of words. The words in each document are randomly drawn from these topics (Blei et al., 2003).

The advantage of the LDA model lies in its unsupervised learning capability, which enables it to automatically uncover latent topics within text without the need for manual annotation. This makes it particularly suitable for processing large-scale textual data, such as policy texts. As a result, LDA has been widely applied in the field of political science (Jelodar et al., 2019). In comparison to models such as BERTopic, LDA demonstrates higher recognition accuracy in long policy documents, with topic identification rates reaching 80% in small corpora and 97% in large ones, compared to 9% and 93% for BERTopic under similar conditions (Kirilenko & Stepchenkova, 2025). Existing studies have applied the LDA clustering model to extract themes from policy texts and identify policy objectives, including analyses in areas such as environmental protection (Davidson et al., 2025), energy development (Oosthuizen & Inglesi-Lotz, 2022), finance (J. Zhang et al., 2024), marine protection (M. Y. Zhu et al., 2023), food safety (Song et al., 2022) and cybersecurity (Chaudhuri et al., 2024).

Prior to modeling, policy texts were preprocessed by removing punctuation, numbers, and special symbols using regular expressions, performing word segmentation with Jieba, and eliminating stopwords based on an extended Chinese stopword list. Term frequency-Inverse document frequency (TF-IDF) weighting was applied to reduce high-frequency noise before converting to term frequency for LDA training. The vectorizer was configured to extract the top 1,000 features, with a maximum document frequency of 0.5 and a minimum document frequency of 2, in order to reduce noise from overly common or rare terms. Vectorization was implemented using the CountVectorizer class, and the fit_transform method was applied to convert the text data

into a sparse term-frequency matrix for TF-IDF weight calculation. For hyperparameter tuning of the LDA model, the number of components (n_components) was treated as the primary parameter. Models were trained sequentially for values ranging from 1 to 15, and stability and repeatability were assessed. The model hyperparameters were set as follows: max_iterations=50, learning_method='batch', learning_offset=50, and random_state=0. After calculating perplexity for each configuration, the optimal number of topics was selected based on a significantly slower rate of perplexity decline and the most informative visualization using the pyLDAvis (Song et al., 2024).

In the LDA model, a document is generated through the following steps: (1) Sample from the dirichlet distribution α to generate the topic distribution θ_i for document i; (2) Sample from the multinomial distribution of θ_i to generate the topic $z_{i,j}$ for the $j-th$ word in document i; (3) Sample from the dirichlet distribution β to generate the term $\phi_{z_{i,j}}$ for the topic $z_{i,j}$; (4) Sample from the multinomial distribution $\phi_{z_{i,j}}$ to produce the term $w_{i,j}$.

The joint distribution of all visible and hidden variables in the model is expressed in Equation 5.

$$p(w_i, z_i, \theta_i, \phi|\alpha, \beta) = \prod_{j=1}^{N} p(\theta_i|\alpha)p(z_{i,j}|\theta_i)p(\phi|\beta)p(w_{i,j}|\phi_{z_{i,j}}). \quad (5)$$

The maximum likelihood estimate for the word distribution of the resulting document can be obtained by summing z_i through the integration of θ_i and ϕ, as shown in Equation 6. Using the maximum likelihood estimate of $p(w_i|\alpha, \beta)$, the model parameters can be estimated using methods such as Gibbs sampling.

$$p(w_i|\alpha, \beta) = \int_{\theta_i} \int_{\phi} \sum_{z_i} p(w_i, z_i, \theta_i, \phi|\alpha, \beta). \quad (6)$$

Once the types of policy instruments and objectives are clearly identified, typical representative policies can be selected to evaluate their effectiveness.

3.3.4 Quantitative model of policy effectiveness

PMC-Index model can be used to assess the consistency level of policy setting and identify the strengths and weaknesses of policy models, which is a standard method for policy evaluation (Estrada, 2011). In contrast to traditional case study methods, the PMC-Index model allows for a more intuitive understanding of policy effectiveness by quantifying the PMC score and visualising the PMC-Surface. Additionally, the PMC-Index model employs a standardised evaluation system,

enabling the comparison of different policies. A series of studies have thoroughly validated the effectiveness of the PMC-Index model (Wang & Xing, 2022; Yang et al., 2021). GLIPs typically involve multiple policy participants and a variety of implementation objectives. Therefore, a multidimensional policy evaluation framework is necessary to simplify the GLIP evaluation process and accurately assess its effects.

The construction of PMC-Index model consists of the following four steps: (1) the use of multi-input-output table; (2) classification of variables and identification of parameters; (3) measurement of the PMC-Index; (4) construction of the PMC-Surface. Fistly, this article constructs a multi input-output table to ensure that all policy texts can be evaluated under the same standards. Secondly, we identified ten key variables drawing on recent research literature, including policy objectives, policy nature, policy timeliness, policy evaluation, policy level, exchange and cooperation, fields involved, industrial collaboration, implementation receptors and policy disclosure. Table S4 presents the definition of the indicators as well as the rationale for their selection. After the first-order variables are determined, this study further defines the second-order variables under each first-order variable and explains them. In policy scoring, the second-order variables are independent. Referring to Estrada's calculation method, this paper adopts the binary assignment method: if the conditions are met, the value of 1 is assigned; if the conditions are not met, the value of 0 is assigned, as shown in Equation 7-8. The paper identified 36 secondary variables (Table 4). Thildly, this paper calculates the PMC-Index for various policies. The PMC-Index score is the sum of scores of the mean values of each sub-variable under the primary variable (Equations 9 to 10). The final evaluation result can be divided into the following four grades: excellent (9 to 10), good (7 to 8.99), qualified (5 to 6.99) and poor (0 to 4.99). Fourthly, the first-order variables obtained in the third step are converted into a 3×3 matrix, from which a three-dimensional PMC-Surface diagram is constructed based on Equation 11. Figure 4 visually illustrates this process. Section 2 of the supplementary materials presents an example illustrating the quantification of the policy using the PMC-Index.

$$X \sim N[0,1], \tag{7}$$

$$X = \{XR: [0,1]\}, \tag{8}$$

$$X_i = \sum_{j=1}^{N(X_{ij})} \frac{X_{ij}}{N(X_{ij})} = \{XR: [0 \sim 1]\}, i = 1,2 \ldots,10. \tag{9}$$

$i = main\ variable;\ j = sub\ variable;\ N(X_{ij})\ is\ the\ total\ number\ of\ the\ sub-variable$

$$PMC = \{X_1 + \cdots + X_m\}, \tag{10}$$

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$$PMC(Y) = \begin{Bmatrix} X_1 & X_2 & X_3 \\ X_4 & X_5 & X_6 \\ X_7 & X_8 & X_9 \end{Bmatrix}. \quad (11)$$

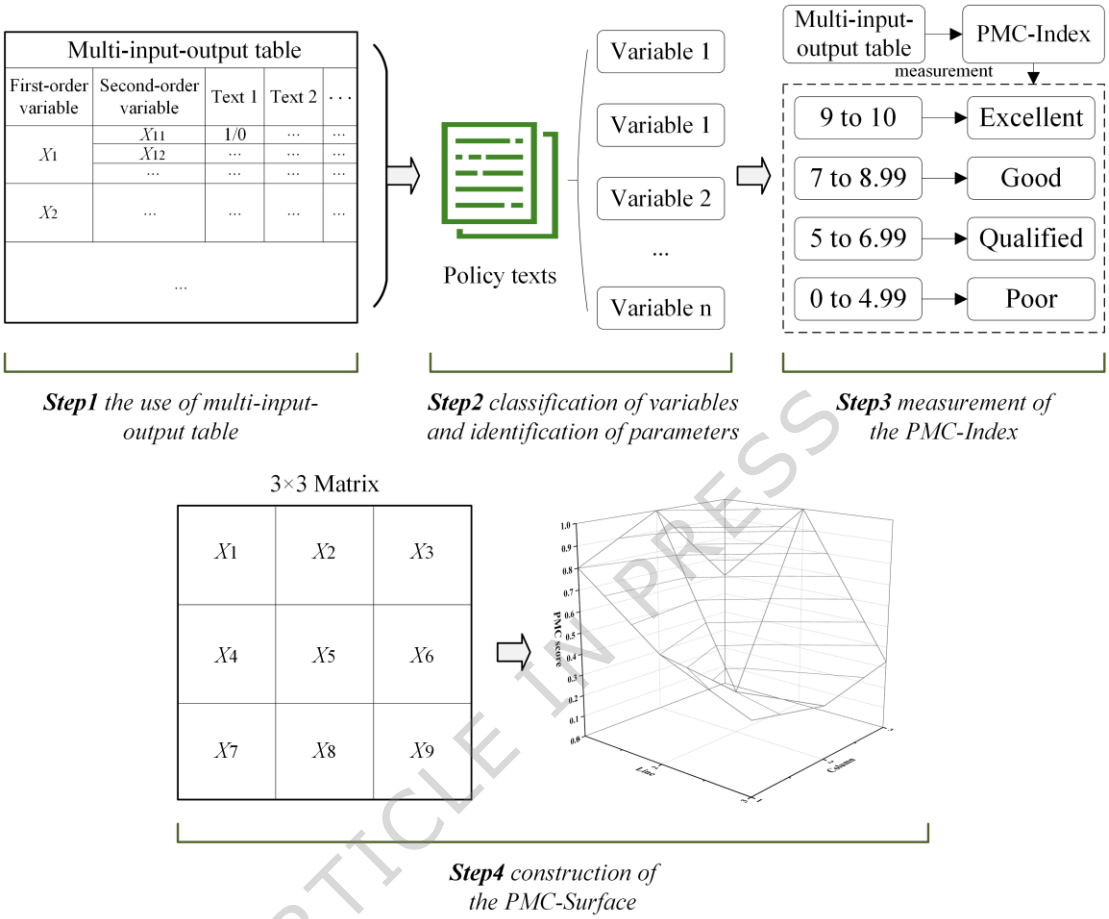


Fig. 4 Flowchart of the PMC-Index model framework

Table 4 Principal variable selection of PMC-Index model

First-order variable	Code	Second-order variable	Code	Second-order variable interpretation
Policy objectives	X_1	Popularisation and application	X_{11}	Whether the policy reflects the view of popularisation and application
		Encourage innovation	X_{12}	Whether the policy reflects the idea of encouraging innovation
		Energy conservation and emission reduction	X_{13}	Whether the policy reflects the view of energy conservation and emission reduction
		Normative guidance	X_{14}	Whether the policy reflects the normative guidance view
		Management constraint	X_{15}	Whether the policy reflects the view of management constraints
Policy	X_2	Guidance	X_{21}	Whether the policy is guiding
		Supervision	X_{22}	Whether the policy is supervisory

nature		Prediction	X_{23}	Whether the policy is predictive
		Recommendation	X_{24}	Whether the policy is recommended
Policy timeliness	X_3	Short term (1 to 3 years)	X_{31}	Whether the policy timeliness includes short-term policies
		Medium term (3 to 5 years)	X_{32}	Whether the policy timeliness includes medium-term policies
		Long term (more than 5 years)	X_{33}	Whether the policy timeliness includes long-term policies
Policy evaluation	X_4	Theoretical basis	X_{41}	Whether the policy involves a theoretical basis
		Practical basis	X_{42}	Whether the policy involves a practical basis
		Clear objectives	X_{43}	Whether the policy involves clear objectives
		Encourage innovation	X_{44}	Whether the policy involves encouraging innovation
Policy level	X_5	Laws and regulations	X_{51}	Whether the policy is a law or regulation
		Administrative regulations	X_{52}	Whether the policy is an administrative regulation
		Departmental regulations	X_{53}	Whether the policy is a departmental regulation
		Normative document	X_{54}	Whether the policy is a normative document
		Industry regulation	X_{55}	Whether the policy is industry regulation
Exchange and cooperation	X_6	Cross-country	X_{61}	Whether the scope of the policy is cross-country
		Trans-regional	X_{62}	Whether the scope of the policy is trans-regional
		Multi-industry	X_{63}	Whether the scope of the policy is multi-industry
		Across periods	X_{64}	Whether the policy exists across periods
Fields involved	X_7	Economy	X_{71}	Whether the policy involves economic growth
		Society	X_{72}	Whether the policy involves social services
		Technology	X_{73}	Whether the policy involves technological innovation
		Politics	X_{74}	Whether the policy has a political content
		Environment	X_{75}	Whether the policy addresses environmental issues
Industrial collaboration	X_8	Industrial chain	X_{81}	Whether the policy has the idea of an industrial chain
		Innovation chain	X_{82}	Whether the policy has the idea of an innovation chain
Implementation receptors	X_9	Government	X_{91}	Whether the policy targets include the government
		Enterprise	X_{92}	Whether the policy object includes enterprises

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		Public	X_{93}	Whether the policy targets include the public
Policy disclosure	X_{10}	None	None	Whether the policy is public

4. Results and analysis

4.1 Quantitative analysis of GLIPs

4.1.1 Reliability testing of policy coding

Two doctoral students were invited to participate in the secondary coding of the thesis. After the relevant coding rules were explained and training was conducted, 102 policy samples were independently coded to assess the reliability of the policy text. The policy text coding consistency coefficient K is found to be 0.8824 (90/102) according to Equations 1 and 2, indicating a high level of agreement between the two coders in their policy coding results. Additionally, the coding reliability R is calculated to be 0.9375, suggesting that the policy text demonstrates high reliability.

4.1.2 Construction of policy analysis framework

This study aims to construct an X-Y two-dimensional analysis framework to better understand the decision-making behaviours and preferences of government departments in policy announcements. The X dimension represents the policy instruments employed by the government, reflecting the methods used to address public issues. The Y dimension represents the policy objectives, outlining the problems the government seeks to solve. To avoid the problem of overfitting and underfitting of clustering topics, we combine SSE, elbow method and pyLDAvis library to determine the number of clusters and then manually define cluster types for explanation.

In K-means clustering, the paper primarily adopts the basic idea of the elbow method. With the increase in cluster number, SSE will continue to decrease until an inflection point is reached, and then the decline speed of SSE will be significantly slowed down. This inflection point is called elbow point, and the cluster number before elbow point is considered the best cluster number. Considering that the number of texts is roughly 20–40, the number of attempts is set to cluster from 1 to 10. According to the generation of inflection points and the judgment of the SSE slope, the demand-side policy instruments are determined to include three types of clustering themes: international cooperation, pilot demonstration and declaration and recommendation. Environmental policy instruments include the following two policy themes: legal provisions and formulation of plans; Supply-side policy instruments include the following three policy themes:

technology application, financial support and public services (Table S5). To evaluate the applicability and effectiveness of the model, we compare the K-means clustering model with the DBSCAN clustering model, another well-established clustering technique. Using the complete policy text from this study as the dataset, we assess model performance based on clustering efficacy and the identification of outliers—texts classified as -1 due to the model's inability to assign them to any category. A higher number of outliers indicates poorer model performance. The number of clusters is set to eight, as shown in Table 5.

Table 5 Model comparison

Model	Clustering results								
	-1	0	1	2	3	4	5	6	7
K-means	0	20	12	2	6	11	23	11	17
DBSCAN	66	16	3	4	5	2	2	2	2

The results indicate that the K-means clustering model outperforms the DBSCAN model in terms of both policy instrument classification accuracy and the identification of outliers. This demonstrates the applicability and superiority of the K-means clustering model.

In LDA clustering, they are relatively repetitive given that elements of some topics in the clustering results have similar meanings to elements in other topics. The clustering results are visualised with the pyLDAvis to achieve a better clustering effect, and further artificial clustering is carried out. Policy texts can be found to be divided into three main themes; that is, when the number of policy clusters is set to 3, the clustering effect is the best. Based on this finding, we finally extract three themes and summarise 60 theme elements which include the elements with high frequency and good performance in this theme but with poor performance in other themes. Then, we continue to manually cluster the policy texts and determine the subject of each policy, as shown in Table 6. The released GLIP objectives of CCEC are divided into the following three categories: Firstly, industrial structure optimisation, which means that the government issues policies to lead the transformation of GLI, deeply integrate new technologies, empower new forms of traditional industries and optimise and upgrade the industrial structure. Secondly, the development of pollution reduction and carbon reduction refers to the government's promulgation of technology-related policies, the use of advanced technological means and management methods to reduce pollutant emissions, reduce carbon emissions and promote the development of low-carbon economy. Thirdly, the emission reduction concept is practiced. The government encourages enterprises and the public to practice the concept of low-carbon life, promote renewable energy

and energy-saving emission reduction technologies, actively participate in environmental protection actions and jointly build a green ecological environment. Through the topic extraction model of LDA text, this study constructs the GLIP objectives chain (Fig. 5).

Table 6 Artificial clustering policy topics based on LDA

Text clustering topic	Topic keywords	Clustering topic	Policy subject	Policy quantity
Topic 1	Green, quantitative, financial bonds, implementation, field, striving, ready mix, green, model, Industrial park	industrial structure optimisation	Chengdu	14
			Chongqing	17
Topic 2	Compliance, computers, construction, batteries, industrial silicon, cleaning agents, concrete, enterprises, assets, industries	pollution and carbon reduction development	Chengdu	22
			Chongqing	10
Topic 3	Service guide, hold tight, counseling, performance, receive, efficient, assessment, industry, solid, insurance law	implementation of emission reduction concepts	Chengdu	18
			Chongqing	21

Fig. 5 shows the construction process and results of the two-dimensional analysis framework.

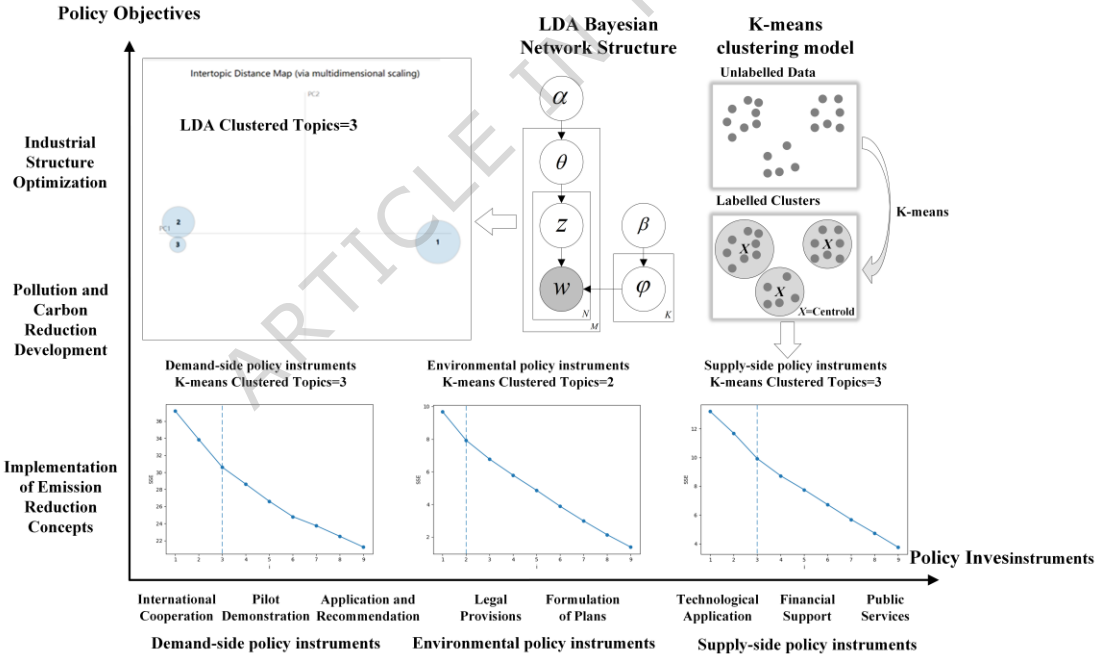


Fig. 5 Construction of policy analysis framework

4.1.3 Distribution of GLIPs in the CCEC

Based on the two-dimensional analysis framework established above, we determine the distribution of GLIPs in the CCEC, as shown in Fig. 6. As can be seen from the figure, all policies are evenly distributed in the two-dimensional framework, without noticeable difference. Thus, the

CCEC area can comprehensively use various policy instruments and formulate diversified multiple policy objectives to promote the development of GLIP.

Notably, the policy instruments covered by the various policy objectives are significantly different. Under the policy objective of industrial structure optimisation, demand-side policy instruments and environmental policy instruments are used more frequently. Most of the policies focus on the application recommendation and formulation of pilot. Under the development policy objectives of pollution and carbon reduction development, the policy focuses on using demand-side policy instruments. Pilot demonstration, technology application and formulation of plans are often used under this policy objective, whilst the policy objective uses less financial support, public services and international cooperation policy instruments. In the subsequent formulation process, emphasis can be placed on the implementation of the concept of pollution reduction in the aspects of economy and publicity to promote the GLI in reducing pollution.

The implementation of emission reduction concepts is the most frequently used policy objective in the CCEC, among which demand-side policy instruments are used the most frequently. By contrast, environmental policy instruments are used the least. Policy makers typically choose international cooperation and public services as policy instruments. Moreover, the policy objective of the technology application frequency is low. Future policy formulation must consider the implementation of technical means to provide the public with a better understanding regarding the emission reduction concept and the formation of a green ecosystem.

Additionally, we observe that certain policy instruments have not been employed to achieve their intended policy objectives, a phenomenon evident in the three policy instruments under consideration.

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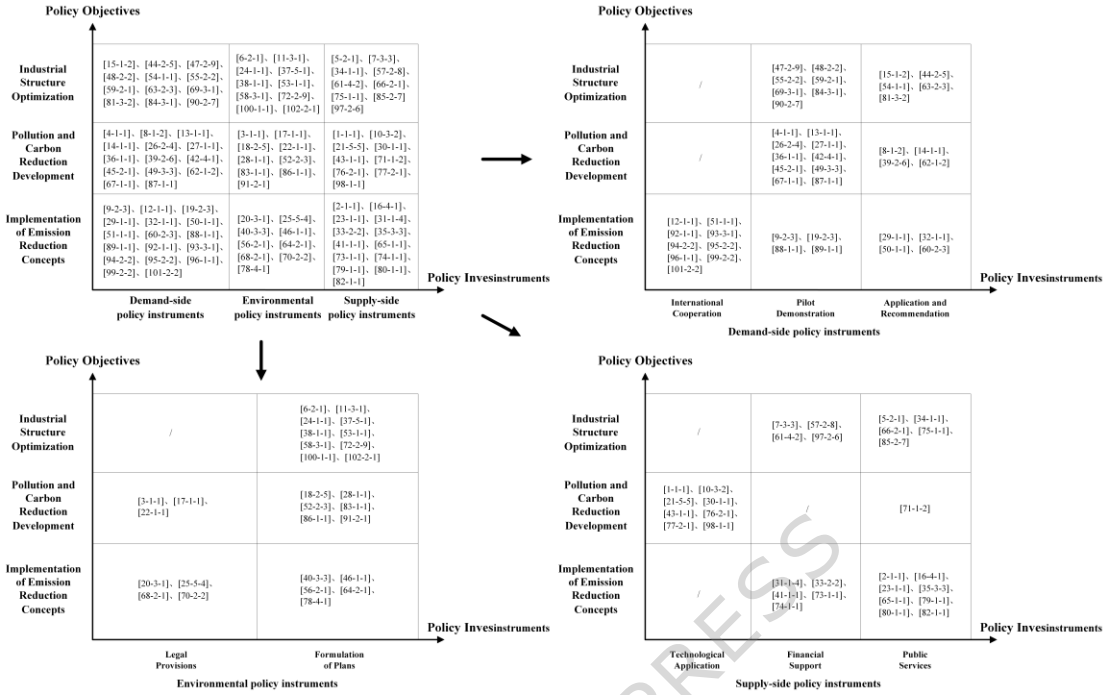


Fig. 6 Two-dimensional distribution of GLIPs in the CCEC

4.1.4 Evolution process of GLIPs

After the descriptive statistical code analysis of policies in the CCEC, we will divide the stages, dig out the critical points of policy texts in different periods and sort out the policies' evolution process. By referring to the overall layout of GLI by the central government in different periods and combining with the time interval of the collected policy texts, the GLIPs in the CCEC are primarily divided into three stages. The first is the Initial Stage of Industrial Policy (2007 to 2019). The idea of the GLI is not widely mentioned. Its related policies stop and punish heavy emissions and heavy pollution behaviours, and its core concept has not been promoted.

The second is the Development Stage of Industrial Policy (2020–2021). In September 2020, China formally proposed achieving a carbon peak by 2030 and carbon neutrality by 2060 at the 75th session of the United Nations General Assembly. In the same year, the National People's Congress and the Chinese People's Political Consultative Conference proposed that the concept of green and low-carbon development has been propagated and promoted, the contradiction between economic development and environmental protection has been reduced day by day and the overall policy principle has shifted from emphasising the economy over the ecology to emphasising the ecology over the economy.

The third is the Transition Stage of Industrial Policy (2022–2023). In 2022, a large number of

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policies to reduce carbon emissions have been introduced. For example, The State Council of the People's Republic of China issued the *14th Five-Year Plan* for energy conservation and emission reduction. The Ministry of Housing and Urban-Rural Development of the People's Republic of China and National Development and Reform Commission of the People's Republic of China issued the *Implementation plan for carbon peaking in urban and rural construction*, among others. For people from all walks of life, staying committed to a green and low-carbon path to development has become the key to development; the concept of green and low-carbon development has been deeply embedded in the government, enterprises and the public and industrial policies and environmental policies have shown high coordination. Although the policy has clear pertinence and effect, some defects remain.

In sum, this section divides GLIPs into three periods, laying the foundation for the subsequent analysis of policy frequency and policy evolution in the time dimension.

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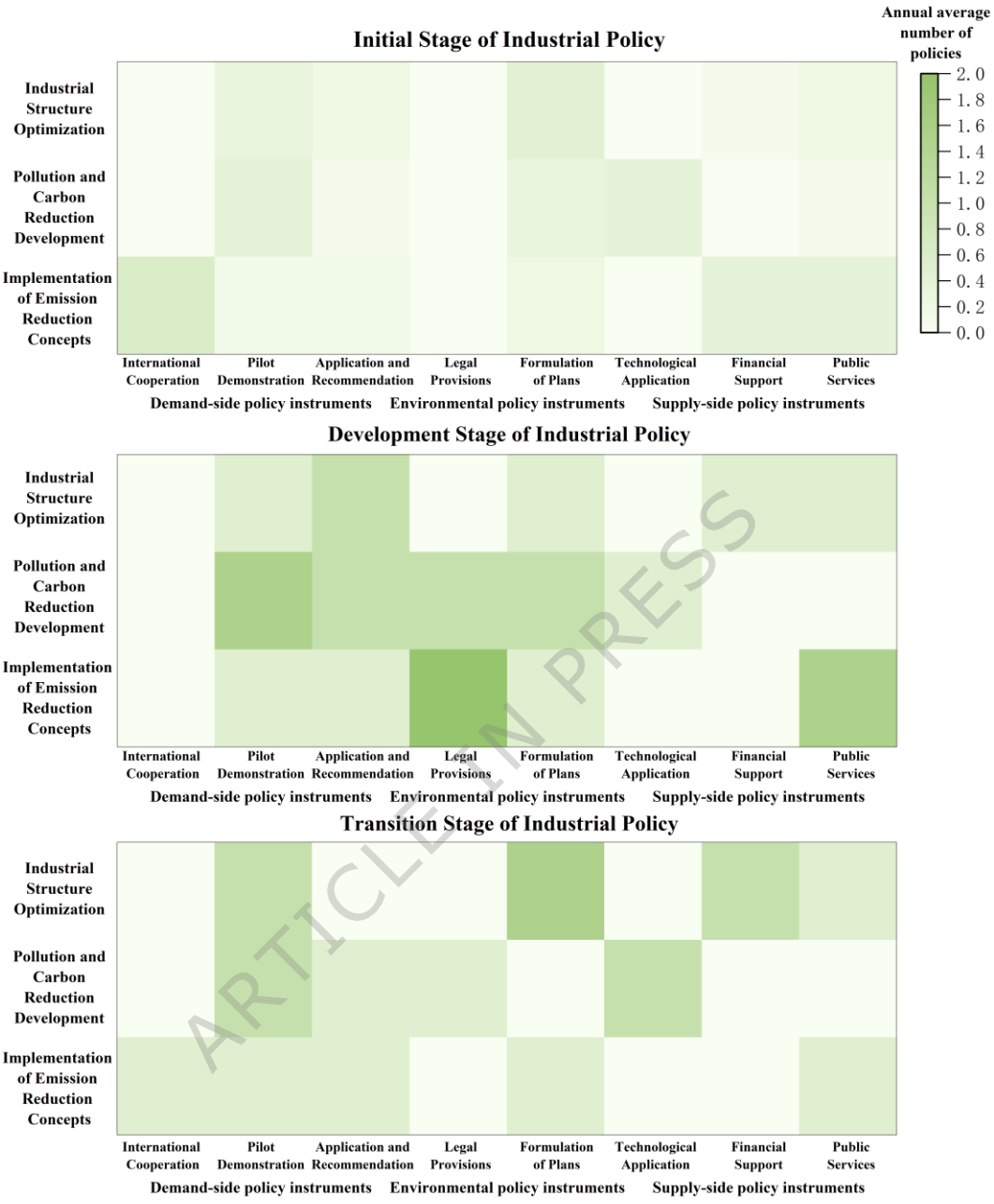


Fig. 7 GLIP evolution heat map

The thermal map can directly show the focus of policy objectives and policy instruments in the three-stage development of the GLI in the CCEC (Fig. 7). The darker the color of the thermal map, the more policies announced by the government on average each year, that is, the more policies focused on the relevant direction. From the perspective of policy instruments, in the three stages, the number of GLIPs in CCEC changed from the majority of demand-side policy instruments to the same number of environmental policy instruments and demand-side policy

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instruments, before finally turning back to the majority of demand-side policy instruments. The policy instruments employed by the CCEC at each stage consistently focus on addressing demand-side needs.

From the perspective of policy objectives, the CCEC initially emphasises the implementation of emission reduction concept. This focus then shifts towards the pollution carbon reduction development, ultimately prioritising the industrial structure optimisation. Overall, a clear trend arises in the GLI policy goals towards industrial structure optimisation. This shift demonstrates a gradual transition in policy focus from abstract concepts to concrete, actionable measures.

In the three stages of policy evolution, government departments formulated different corresponding policies according to the industrial development course of the CCEC. In the initial stage of industrial policy (2007–2019), the policy's main objective was to promote economic growth in the CCEC, especially in the rapid development of traditional manufacturing and emerging industries. At this stage, policymakers mostly promote enterprise investment through financial support and technology application, encourage technological innovation and industrial upgrading to solve supply-side structural problems and enhance industrial competitiveness. The policy also focuses on expanding market demand, promoting the growth of domestic demand and improving the vitality of the regional economy through pilot and demonstration policies. In the development stage of industrial policy (2020–2021), with the gradual transformation of the economic structure in the CCEC, environmental protection and sustainable development have become an essential focus of policy formulation. The industry in the area has steadily transformed from high energy consumption and high pollution industries to green industries and high value-added industries. In the transformation stage of industrial policy (2022–2023), the industrial policy in the CCEC has entered a diversified and refined stage, and policymakers are flexible in using various policy instruments. On the basis of the previous stage, the subject of policy implementation no longer relies on a single policy instrument or policy objective but can combine various policy instruments to achieve the comprehensiveness and coordination of policies. Through financial instruments, scientific and technological support and regional integration development strategy, policy implementers promote inter-regional cooperation.

We identified three distinct color blocks in the analysis framework, which we have labeled as the policy focus. These blocks indicate the clear tendencies in the policy objectives and decision-making behaviours of government departments at each stage. In the Development Stage of

Industrial Policy, the government's primary focus is on implementing emission reduction strategies. To address this concern, government departments prioritise the development of environmental policy instruments and the enhancement of public service infrastructure (supply-side policy instruments). The second block appears in the Transition Stage of Industrial Policy, where the government shifts its focus towards adjusting the industrial structure. In this stage, the government aims to establish various planning guidelines to address this issue through a directive policy approach.

4.2 GLIP effectiveness evaluation and analysis

4.2.1 Calculation and evaluation of PMC-Index

The PMC-Index model is a model-based approach to evaluate public policy setting consistency. Based on relevant theories and empirical research, this method constructs a multidimensional index system to evaluate and analyse the published policies to demonstrate the effectiveness of the policies quantitatively. In the selection of GLIP, to add theoretical and analytical robustness to the policy texts and feasibility of the results, the paper heeds to the following principles: (1) The selected policy texts are representative, forward looking and directional; in addition, the text content can influence the future transformation and upgrading trend of GLI and lead the industrial development and (2) The selected policy text includes all three types of policy instruments to comprehensively assess their effectiveness; (3) The selected texts are time-sensitive, and the collected policy texts have a large period from 2007 to 2023. The article chooses the policies issued in recent years. Additionally, its content can reflect the characteristics of the current era, and its practical application is robust. In combination with the above regulations, four policies were selected as the policy evaluation objects for PMC-Index analysis, as shown in Table 7. The four policies not only have representativeness, relevance and timeliness; their release time is also in the alternating period of the evolution of GLI. Furthermore, their theoretical significance and practical application value are prominent with analytical relevance.

Table 7 Representative policy selection

Policy name	Policy number
Notice of Sichuan Provincial Department of Economy and Information Technology and Sichuan Provincial Department of Finance on the implementation of the Provincial Government's 'Several Policies to Support the High-quality Development of Green and Low-carbon Advantageous Industries'.	S_1
Guidance of State-owned Assets Supervision and Administration Commission of Sichuan Provincial Government on Carbon Peak carbon neutrality of provincial enterprises	S_2

Notice of Chongqing Municipal People's Government on issuing the Action Plan on Further Promoting High-quality Green Development of Manufacturing Industry (2022–2025) with the goal of achieving carbon Peak carbon neutrality as the guide	C_1
Notice of Chongqing Housing and Urban-Rural Development Commission, Chongqing Operation Management Department of the People's Bank of China and Chongqing Supervision Bureau of the China Banking and Insurance Regulatory Commission on the issuance of the Pilot Plan for the Coordinated Development of Green Building Industry and Green Finance in Chongqing	C_2

According to Equation 7, the policies were scored, and the sub-dimensions of PMC-Index scores of the four policies were obtained, as shown in Table 8. To ensure the robustness and reliability of the results, a professor and a doctoral student in the same field independently replicated the assignment calculations. As shown in Table S6, their results closely align with those reported in this study, and the final score rankings of the four policies remain unchanged, thereby demonstrating the robustness of the model.

Table 8 PMC sub-dimension scores

First-order variable	Second-order variable	S_1	S_2	C_1	C_2
X_1	X_{11}	1	1	1	0
	X_{12}	1	1	1	1
	X_{13}	1	1	1	1
	X_{14}	0	1	1	1
	X_{15}	1	0	0	1
X_2	X_{21}	1	1	1	1
	X_{22}	1	0	0	1
	X_{23}	0	0	1	0
	X_{24}	0	1	1	1
X_3	X_{31}	1	0	0	1
	X_{32}	0	1	1	1
	X_{33}	0	1	1	0
X_4	X_{41}	1	1	1	1
	X_{42}	1	0	1	0
	X_{43}	1	1	1	1
	X_{44}	1	1	1	1
X_5	X_{51}	0	0	0	0
	X_{52}	0	0	0	0
	X_{53}	0	0	1	1
	X_{54}	1	1	1	1
	X_{55}	0	0	0	1
X_6	X_{61}	0	0	0	0
	X_{62}	0	1	1	0
	X_{63}	1	1	1	1
	X_{64}	0	0	1	0
X_7	X_{71}	1	1	1	1

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	X_{72}	0	1	1	1
	X_{73}	1	0	1	0
	X_{74}	0	0	1	0
	X_{75}	1	1	1	1
X_8	X_{81}	1	0	1	1
	X_{82}	1	1	1	1
X_9	X_{91}	0	0	1	0
	X_{92}	1	1	1	1
	X_{93}	0	1	1	1
X_{10}	\	1	1	1	1

Equation 10 is modified, and Table 9 shows the calculation results of PMC scores for each policy.

Table 9 PMC-Index score

Policy	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	Score
S_1	0.8	0.5	0.33	1	0.2	0.25	0.6	1	0.33	1	6.01
S_2	0.8	0.5	0.67	0.75	0.2	0.5	0.6	0.5	0.67	1	6.19
C_1	0.8	0.75	0.67	1	0.4	0.75	1	1	1	1	8.37
C_2	0.8	0.75	0.67	0.75	0.6	0.25	0.6	1	0.67	1	7.09
Mean value	0.8	0.625	0.585	0.875	0.35	0.4375	0.7	0.875	0.6675	1	6.915

We use spiderwebs to visualise the scores of each of the policy's first-order variables. As can be seen from Fig. 8, C_1 has a high score in the relevant fields, policy transparency, implementation recipients, industry coordination and policy evaluation, which is one of the reasons why it ranks first in the final score. S_1 scores poorly in the implementation recipients, policy timeliness and communication and cooperation, and thus ranks fourth.

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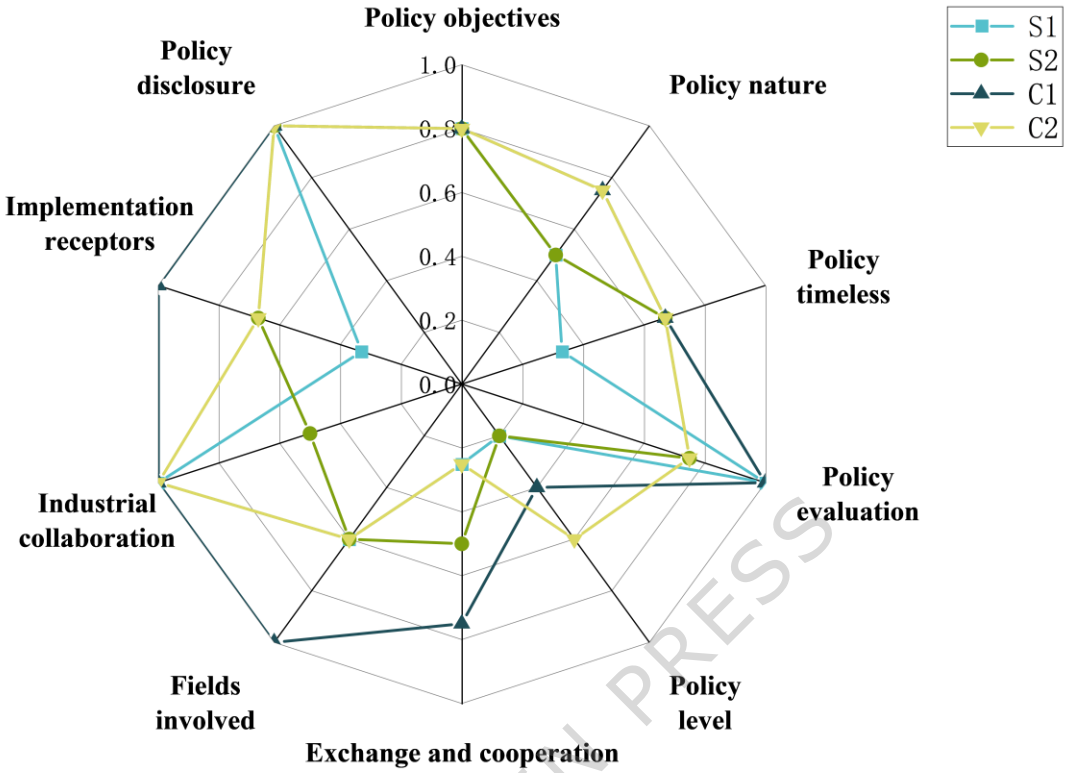
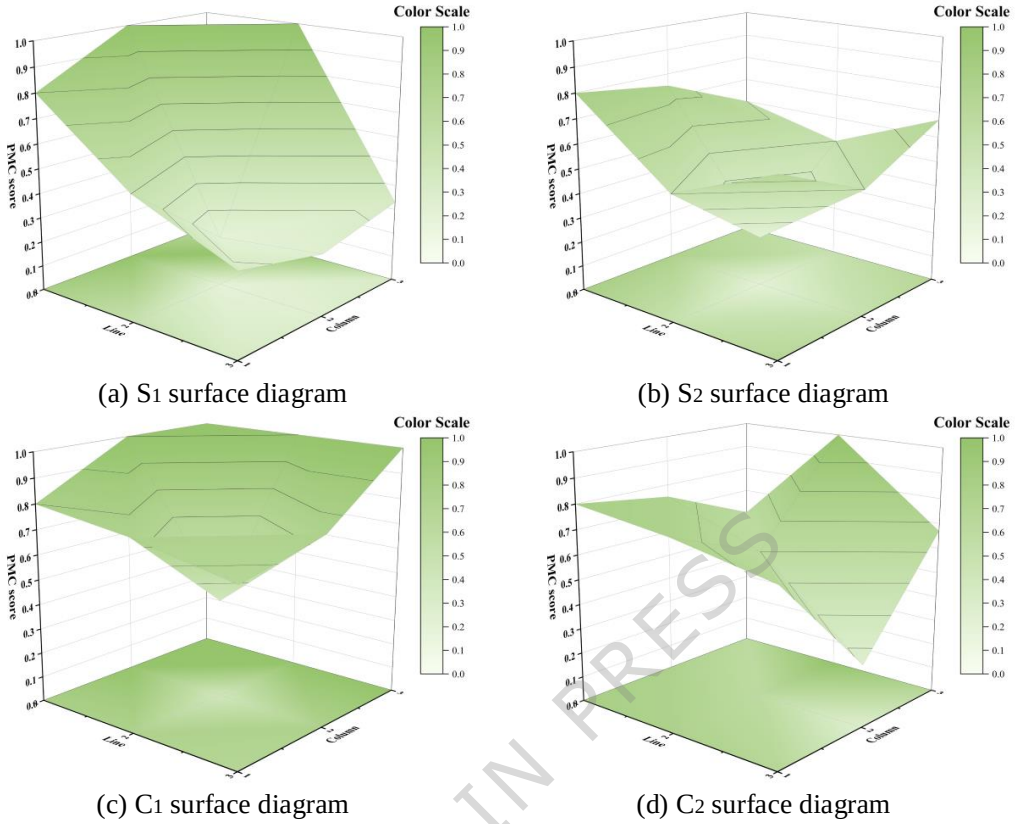


Fig. 8 Score of first-order variables

4.2.2 Construction and plotting of PMC-Surface

After calculating the PMC-Index and judging the evaluation level, we constructed the PMC-Surface map. On the basis of a spider diagram, the PMC-Surface can further display intuitively the PMC-Index results in the form of graphics and further evaluate and analyse the policy effectiveness. We draw a three-dimensional surface diagram of the above policy to analyse its inherent shortcomings by Equation 11, as shown in Fig. 9.

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(a) S1 surface diagram (b) S2 surface diagram

(c) C1 surface diagram (d) C2 surface diagram

Fig. 9 PMC-Surface of GLIPs

According to Fig. 9(a), S_1 is found to be prominent in policy evaluation and industrial synergy but lacks in policy rank and exchange and cooperation. When the policy subject promulgates the subsequent policy, it needs to start the construction of the policy level and exchange and cooperation.

S_2 is relatively balanced as a whole, as can be seen from Fig. 9(b). However, the policy level score is the lowest (0.2). This finding shows the inadequate connection between policies and laws and regulations. In policy formulation and policy revision, policy subjects need to focus on how to combine laws and regulations, administrative regulations and issued policies. In addition, the implementation receptor score is low, indicating that the implementation object of the policy needs to be considered from multiple aspects to ensure that the policy can cover the vast majority of industry stakeholders.

The PMC three-dimensional surface plot of C_1 is similar to the surface plot of S_2. In Fig. 9(c), the score of this policy is not high in the policy level, but it is also higher than that of S_2. In addition, the policy needs to be further strengthened in terms of policy timeliness.

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As can be seen from Fig. 9(d), the overall score of C_2 is high, the score of communication and cooperation is low and the scores of other first-order variables are above 0.5. When the policy subject issues the policy in the future, one must pay attention to the communication and synergy of the policy, promote field exchanges, expand enterprise cooperation and promote the sustainable development of GLI.

5. Discussions

This study proposes a comprehensive model integrating machine learning and the PMC-Index to quantify the role and effectiveness of policy text identification. This study enhances the application of text mining models in policy analysis and deepens the understanding of industrial sustainable development policies in urban agglomerations from a macro perspective. The findings indicate that, whereas the number of policy objectives remains consistent, the policy instruments employed vary significantly. This finding suggests that the CCEC is capable of planning the development of GLI at multiple levels, despite the discrepancies in the regulatory measures emphasized during the implementation process. Regarding the overall use of policy instruments, although demand-side instruments tend to generate short-term, one-off stimulus effects and have a brief policy lifecycle, they remain the most frequently utilised by CCEC governments across various stages of industrial development, reflecting their policy-making preferences.

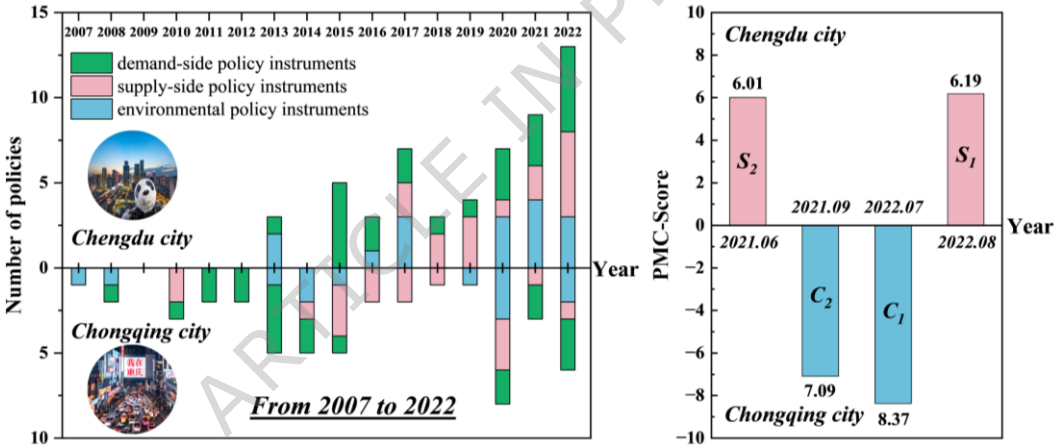
Interestingly, demand-side policy instruments, which serve as stimulants for industrial development, are typically employed by governments only in the early stages of industrialisation (Abbott & Fitjar, 2025). By contrast, other regions and cities tend to favour supply-side policy instruments as their industries mature. For example, the European Union has adopted a combination of regulatory constraints (environmental policy instruments) and market incentives (supply-side policy instruments) in its environmental policies, such as the carbon emissions trading system (Jevnaker & Wettestad, 2017) and renewable energy subsidies (Özdemir et al., 2020). However, the CCEC government consistently prefers demand-side policy instruments throughout all stages of industrial development. One possible explanation for this phenomenon is that industrial development in China's policy environment largely depends on direct government intervention in the market rather than the organic growth of the market itself (Jia et al., 2020). Additionally, the current GLI development in CCEC may not yet meet the criteria necessary for the long-term use of supply-side or environmental policy instruments. From Figure 7, none of the

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policy instrument dimensions within CCEC's three policy centers fall under demand-side policy instruments. Although demand-side policy instruments account for the largest proportion overall, their distribution is more evenly spread across different stages. By contrast, legal provisions (environmental policy instruments), planning frameworks (environmental policy instruments) and public services (supply-side policy instruments) exhibit a clear clustering effect under specific policy objectives. This finding suggests that at critical stages of industrial development, the CCEC government tends to prioritise long-term, structural adjustment policies. CCEC follows a strategy where demand-side instruments address broad issues, whereas the other two types of instruments focus on specific concerns—similar to strategies used in other developing countries or regions (Kemp & Never, 2017). These findings further support the earlier discussion on the relevance of supply-side and environmental policy instruments in GLIP. The conclusions imply that the policy-making process in CCEC may be influenced by complex factors. Additionally, our analysis of GLIP's policy objectives reveals that CCEC's policy guidance is highly effective. Relevant policies are formulated on the basis of the industrial development life cycle, and these policies exert a continuous impact (Oosthuizen & Inglesi-Lotz, 2022). However, despite the strong policy alignment, the long-term effects of these policies essentially require continuous evaluation to prevent imbalances in the policy system's stability (Hildén, 2011).

Another important finding to discuss is our assessment of policy-making in Chengdu and Chongqing, which revealed imbalances both in the overall situation and across certain dimensions of policy making. There are significant differences between Chengdu and Chongqing regarding the frequency and effectiveness of GLIPs (Fig. 10). Overall, Chengdu has only recently begun to focus on the formulation and promotion of GLIPs, resulting in relatively low policy effectiveness. In contrast, Chongqing began emphasizing GLIP releases between 2013 and 2015, and after years of refinement, its policy impact has strengthened. From the perspective of agglomeration economics (Zheng & He, 2022), Chongqing's earlier policy initiatives may have generated cumulative advantages, attracting more resources and institutional attention, whereas Chengdu's development has progressed more slowly. These disparities may also reflect potential siphoning effects and agglomeration shadows in politically polycentric regions (Wang, 2020). Specifically, Chongqing has exhibited stronger industrial clustering in energy-intensive and heavy manufacturing sectors, which may have generated greater institutional pressure and incentives for the early adoption of GLIPs. In contrast, Chengdu's economy is more service- and technology-

oriented, with dispersed industrial nodes, potentially diminishing the immediate urgency of policy implementation and resulting in a slower rollout. Regional institutionalism further suggests that distinct administrative traditions and governance networks in the two cities have influenced their policy-making pace and effectiveness (Gallego et al., 2017). For instance, Chongqing, as a municipality directly under the Central Government of China, possesses stronger vertical coordination capabilities and centralized control over industrial planning, enabling swift policy adoption and consistent enforcement. In contrast, Chengdu relies more on decentralized governance and multi-level stakeholder consultations, resulting in a more deliberative but slower policy formulation process. In practice, differences in local government–industry coordination, the availability of technical expertise, and pre-existing industrial infrastructure appear to reinforce these theoretical expectations. Although Chengdu initially lagged in policy absorption and localization, over time both the quantity and effectiveness of its policies have steadily increased, potentially due to the spatial diffusion of policies (Mitchell, 2018).



(a) Number of policies on Chengdu and Chongqing (b) Evaluation of Policy effectiveness

Fig. 10. Comparative policy announcement timelines of Chengdu and Chongqing

To address these issues, CCEC must enhance regional coordination and optimise the alignment of policy instruments to reduce regional disparities in policy effectiveness. Furthermore, policy rank is the most significant negative factor influencing the PMC-Index score. A series of policies are designed and implemented at the top level to ensure that macro strategies align with local implementation capacity, providing guidance and support for regional industry development (Zang et al., 2025).

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6. Conclusions

6.1 Conclusions and suggestion

The study collected and analysed 102 GLIPs from CCEC, totaling 188,934 characters. K-means and LDA text mining models were applied to evaluate the phase characteristics of GLIPs, whilst the PMC-Index was utilised to quantify the effectiveness of representative policies. The key findings of this paper are summarised as follows:

(1) Demand-side policy instruments are the most frequently used policy instruments by CCEC (43 instances, 42.16%), a trend observed across all three stages of policy development discussed in this study. These figures are followed by supply-side policy instruments (31 instances, 30.39%) and environmental policy instruments (28 instances, 27.45%). Among the three types of policy objectives, the implementation of the emission reduction concept encompasses the largest number of policy objectives (39 instances, 38.24%), followed by pollution and carbon reduction development (32 instances, 31.37%) and industrial structure optimisation (31 instances, 30.39%). Notably, despite the similar number of policy objectives, the frequency of policy instrument usage varies significantly. Certain policy instruments are found to address only one specific policy objective (e.g., international cooperation is solely associated with the implementation of emission reduction concept). This finding suggests potential limitations in policy instruments and a governmental reliance on these instruments.

(2) The GLIP formulation by CCEC spans three stages: the initial stage of industrial policy (2007–2019), the development stage of industrial policy (2020–2021) and the transformation stage of industrial policy (2022–2023). Our analysis shows that CCEC consistently favoured demand-side policy instruments across all stages. However, the policy objectives evolved significantly—from focusing on emission reduction concepts to prioritising pollution and carbon reduction, ultimately shifting to industrial structure optimisation. This shift indicates that CCEC's policymaking is increasingly focused on specific control measures, moving beyond theoretical proposals towards industrial restructuring. This pattern reflects the general trajectory of industrial development, which typically follows three stages: exploration, control and transformation. Indeed, policy issuance adapts to and guides industrial development. Despite the previous emphasis on demand-side policy instruments as the standard approach for addressing policy objectives, none of the key policies identified by the analysis framework employ demand-side policy instruments.

This discrepancy may be attributed to China's unique policy environment.

(3) The PMC policy score for Chongqing is significantly higher than that of Chengdu, with a score gap of 1.38, indicating that despite the strength in the overall effectiveness of policy formulation in Chongqing, concerns persist regarding the stability of policy formulation. Furthermore, the PMC-Surface analysis reveals that, across all variables, policy-level scores are low; whereas scores for policy evaluation and industry coordination are high. The surface displays a concave shape in the center and a convex, funnel-like shape on the periphery.

Based on the above findings, this paper proposes the following three recommendations:

(1) In future policy formulation and optimisation, CCEC government entities should prioritise the increased use of environmental and supply-side policy instruments to create a comprehensive policy toolkit. Although demand-side instruments have been widely implemented, their effectiveness is limited when not integrated with broader structural reforms. Prolonged reliance on these instruments can generate fiscal pressures from sustained subsidies, foster industrial dependency that discourages innovation and technological advancement, and increase vulnerability to external market fluctuations. Furthermore, when demand-side measures dominate without complementary supply-side or environmental interventions, policy coordination weakens, leading to fragmented implementation and reduced long-term effectiveness. Additionally, the government should enhance the collaborative design of policy instruments, expand their scope, improve their flexibility and adaptability and reduce reliance on a single policy instrument.

(2) The government should align policy formulation with the industrial development cycle and proactively plan for future stages. Given that the policy goal has shifted towards optimising industrial structure, subsequent policies should focus on identifying the direction of industrial development and crafting targeted, favourable policies. For instance, CCEC could initially pilot policies that promote the clustering of low-carbon industries or the upgrading of traditional manufacturing sectors within specific industrial parks, and subsequently evaluate the outcomes in terms of technological adoption, emission reduction, and economic returns. This scenario will foster a positive feedback loop between the economy and ecology.

(3) Government departments should institutionalize regular mechanisms for policy evaluation and adjustment, rather than relying on ad hoc reviews. Retrospective analyses should be integrated with risk assessments to anticipate unintended consequences, including policy overlap, enforcement gaps, or excessive reliance on short-term incentives. On the one hand, the release of

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a series of policies will improve continuity and coordination, ensuring alignment between policy objectives and effective implementation. On the other hand, such a process will help reduce uncertainty caused by policy fluctuations and promote the stable development of the industry. Additionally, CCEC could consider raising the level of core policy issuance, transitioning from local government policies to provincial or national policies to enhance the authority and effectiveness of policy implementation. A practical approach is to implement a pilot-to-normative pathway, whereby innovative policies are initially trialed at the local level, evaluated for effectiveness, and subsequently scaled into provincial frameworks and, if successful, incorporated into national guidelines. Alternatively, a multi-level policy alignment mechanism can be established, in which municipal policies are designed from the outset to align with provincial strategies and national priorities, facilitating smoother upward transfer and more rapid formal adoption at higher levels of governance.

6.2 Research outlook

This paper has conducted a rigorous quantitative analysis of CCEC's GLIP, with an aim to address the 'black box' issue in the policy formulation and implementation process. The policy evaluation methodology employed in this study has not been previously applied to GLIP, offering potential avenues for new research and providing a theoretical foundation for future evaluations of GLIPs and related energy policies.

However, this study has certain limitations due to constraints in time, scope and technology. Firstly, based on the existing case analysis, future research could incorporate the timing of policy release or policy effectiveness as additional dimensional variables, creating an X-Y-Z three-dimensional policy analysis framework to enable a comprehensive and objective policy evaluation. Secondly, future studies could compare relevant national and local policies, analysing the similarities and differences across various research scales to offer targeted policy recommendations. Lastly, subsequent research may consider the use of simulation modeling to complement and support the assessment of policy implementation outcomes (Sarkar, Kweon, et al., 2024).

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Data availability and access

The data that support the findings of this study are available from the corresponding author, Chengdu University of Technology, upon reasonable request.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

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